WORKED SOLUTION

# STEP II 2002, Question 3

Pure • Fermat Numbers and Primes

**STEP II 2002, Question 3**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q3

This question has a full guided treatment in *Beyond STEP*, Section 1.1, *A Prime Factory*: https://beyondstep.keithmcnulty.org/ch01/fermat-numbers.html

## The key idea

Write $F_k - 2 = 2^{2^k} - 1$. Multiplying this by $F_k = 2^{2^k} + 1$ gives a difference of two squares, and that is exactly the step the induction needs. Once the product formula is in place, any common factor of two Fermat numbers has to divide 2. Since every Fermat number is odd, that factor can only be 1.

## Solution

### The first four Fermat numbers

Since $2^0 = 1$, $2^1 = 2$, $2^2 = 4$ and $2^3 = 8$,

$$F_0 = 2^1 + 1 = 3, \qquad F_1 = 2^2 + 1 = 5, \qquad F_2 = 2^4 + 1 = 17, \qquad F_3 = 2^8 + 1 = 257.$$

Now we check $(*)$ for the first three values of $k$:

$$\begin{aligned} k = 1: &\quad F_0 = 3 = 5 - 2 = F_1 - 2, \\ k = 2: &\quad F_0 F_1 = 15 = 17 - 2 = F_2 - 2, \\ k = 3: &\quad F_0 F_1 F_2 = 15 \times 17 = 255 = 257 - 2 = F_3 - 2. \end{aligned}$$

### Proof of $(*)$ for all $k \geq 1$

We use induction on $k$. The case $k = 1$ is done above.

Suppose $(*)$ holds for some $k \geq 1$, so that $F_0F_1 \cdots F_{k-1} = F_k - 2$. Then

$$\begin{aligned} F_0F_1 \cdots F_{k-1}F_k &= (F_k - 2)\, F_k \\ &= \left(2^{2^k} - 1\right)\left(2^{2^k} + 1\right) \\ &= \left(2^{2^k}\right)^2 - 1 \\ &= 2^{2 \cdot 2^k} - 1 = 2^{2^{k+1}} - 1 \\ &= F_{k+1} - 2. \end{aligned}$$

So $(*)$ holds for $k+1$ as well, and by induction it holds for every $k \geq 1$.

**Ask yourself** Why is $\left(2^{2^k}\right)^2$ equal to $2^{2^{k+1}}$, and not $2^{2^{2k}}$ or something else?

Squaring a power doubles the exponent: $\left(2^a\right)^2 = 2^{2a}$. Here $a = 2^k$, and $2a = 2 \cdot 2^k = 2^{k+1}$. Taking care over towers of exponents like this is most of what the induction step involves.

#### No two Fermat numbers share a factor

Take two different Fermat numbers $F_m$ and $F_n$ with $m < n$. Then $n \geq 1$, and $F_m$ is one of the factors $F_0, \ldots, F_{n-1}$ in $(*)$ with $k = n$. So

$$F_n - 2 = F_m \times (\text{an integer}),$$

which means $F_m$ divides $F_n - 2$.

Now suppose a positive integer $d$ divides both $F_m$ and $F_n$. Since $d \mid F_m$ and $F_m \mid F_n - 2$, we have $d \mid F_n - 2$. Since $d$ divides $F_n$ as well, it divides the difference:

$$d \mid F_n - (F_n - 2) = 2,$$

so $d = 1$ or $d = 2$. But $F_m = 2^{2^m} + 1$ is odd, being an even number plus 1 (note $2^m \geq 1$). So $d \neq 2$, and therefore $d = 1$. Hence no two Fermat numbers have a common factor greater than 1.

#### Infinitely many primes

Each $F_n$ is at least 3, so it has at least one prime factor. Choose one and call it $p_n$. This gives a prime for each $n = 0, 1, 2, \ldots$.

**Ask yourself** Couldn't two different Fermat numbers give the same prime?

No. If $p_m = p_n$ with $m \neq n$, that prime would divide both $F_m$ and $F_n$. It would then be a common factor greater than 1, which the previous part rules out. So the primes $p_0, p_1, p_2, \ldots$ are all different. Since there are infinitely many Fermat numbers, there are infinitely many distinct primes.

Notice that we never needed $F_n$ itself to be prime, only that each one has *some* prime factor that no other Fermat number shares.

## Checks

- $(*)$ with $k = 4$: $255 \times 257 = 65535 = 65537 - 2 = F_4 - 2$. We also checked $(*)$ exactly by computer for all $k$ up to 12.
- The smallest prime factors of $F_0, \ldots, F_5$ are $3, 5, 17, 257, 65537, 641$, all different, just as the argument predicts. $F_5 = 4294967297 = 641 \times 6700417$ is the first Fermat number that isn't prime.
- We confirmed numerically that $\gcd(F_m, F_n) = 1$ for all $0 \leq m < n \leq 10$.

## Common slips

- Reading $2^{2^n}$ as $(2^2)^n = 4^n$. The exponent is $2^n$, so $F_3 = 2^8 + 1$, not $4^3 + 1$.
- Getting $d \mid 2$ and then forgetting to rule out $d = 2$. The fact that Fermat numbers are odd is what finishes the argument.
- Assuming every Fermat number is prime. The final part works without this, and it is false anyway ($641 \mid F_5$).

## Beyond STEP

This proof is due to Pólya. It belongs to a family of proofs of the infinitude of primes. Each builds an infinite sequence of pairwise coprime integers, and each term then contributes its own new prime. Euclid's proof can be read the same way. Fermat numbers come up again in number theory courses: Gauss showed that a regular $n$-gon can be constructed with ruler and compass exactly when $n$ is a power of 2 times a product of distinct Fermat primes, and no Fermat prime beyond $F_4$ is known. For more, read Davenport, *The Higher Arithmetic*.

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